

Question ID: e5c57163

| Assessment | Test | Domain                    | Skill           | Difficulty                               |
|------------|------|---------------------------|-----------------|------------------------------------------|
| SAT        | Math | Geometry and Trigonometry | Area and volume | Hard <div></div> <div></div> <div></div> |

Question

Square A has side lengths that are **166** times the side lengths of square B. The area of square A is  **$k$**  times the area of square B. What is the value of  **$k$** ?

Correct Answer: 27556

Rationale

The correct answer is **27,556**. The area of a square is  **$s^2$** , where  **$s$**  is the side length of the square. Let  **$x$**  represent the length of each side of square B. Substituting  **$x$**  for  **$s$**  in  **$s^2$**  yields  **$x^2$** . It follows that the area of square B is  **$x^2$** . It's given that square A has side lengths that are **166** times the side lengths of square B. Since  **$x$**  represents the length of each side of square B, the length of each side of square A can be represented by the expression  **$166x$** . It follows that the area of square A is  **$(166x)^2$** , or  **$27,556x^2$** . It's given that the area of square A is  **$k$**  times the area of square B. Since the area of square A is equal to  **$27,556x^2$** , and the area of square B is equal to  **$x^2$** , an equation representing the given statement is  **$27,556x^2 = kx^2$** . Since  **$x$**  represents the length of each side of square B, the value of  **$x$**  must be positive. Therefore, the value of  **$x^2$**  is also positive, so it does not equal **0**. Dividing by  **$x^2$**  on both sides of the equation  **$27,556x^2 = kx^2$**  yields  **$27,556 = k$** . Therefore, the value of  **$k$**  is **27,556**.